



Discrete Structures

- Chapter 5, Induction
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 - Spring 2026
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Proof

Proof: an argument that convincingly demonstrate truth of a statement

In Computer Science and Engineering

- Prove that the algorithm is correct
- Prove that algorithm has a particular running time
- Data structure proofs often lead to efficient and simpler algorithm

In general, working with proofs develops useful habits in thinking

- Working with precise notions
- Exactly formulating statements
- Paying attention to all possibilities

Induction

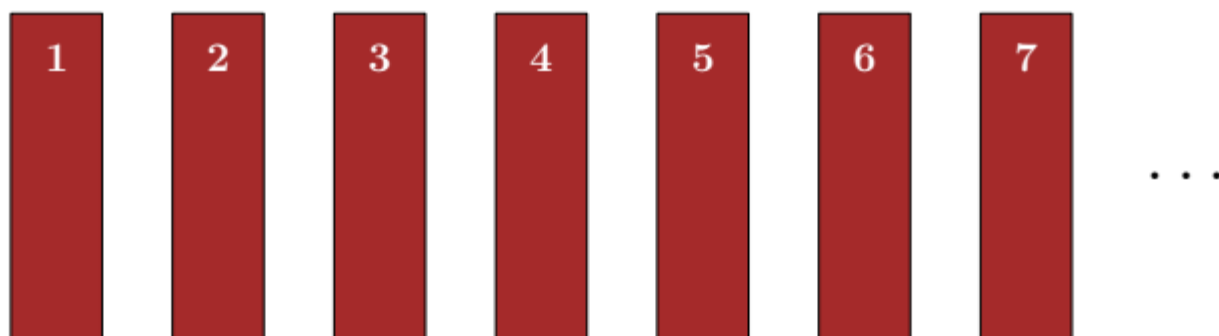
Induction is useful to prove statements about all positive integers,

$$\sum_{i=0}^n i = 1 + 2 + 3 + \dots + (n-1) + n = \frac{n(n+1)}{2}$$

Number of subsets of an n -element set is 2^n

Principle of Induction

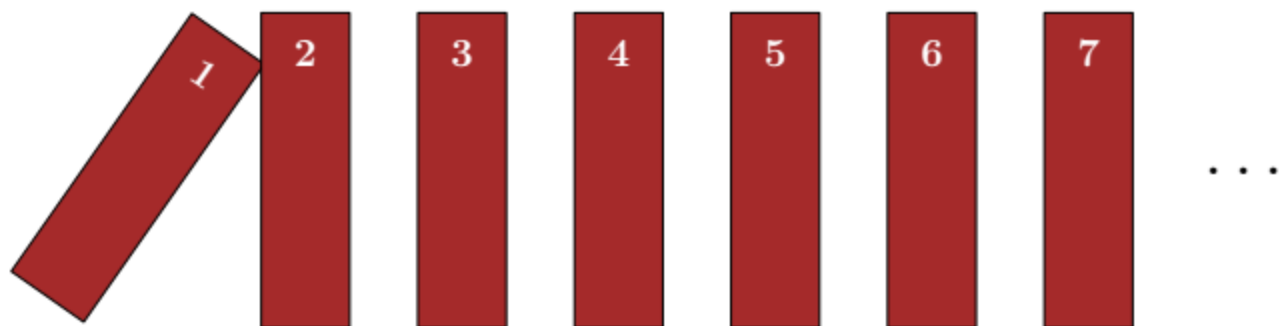
Suppose we have an infinite number of dominoes



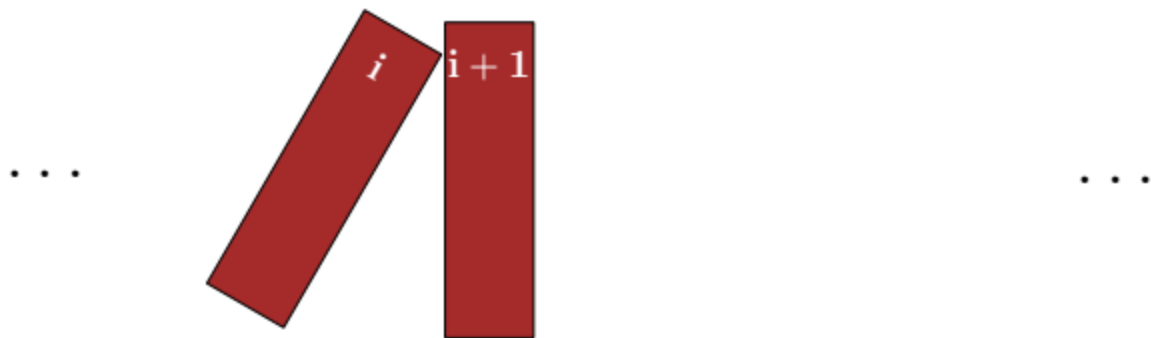
Principle of Induction

Suppose two rules/facts are given

- **Fact 1:** First domino falls



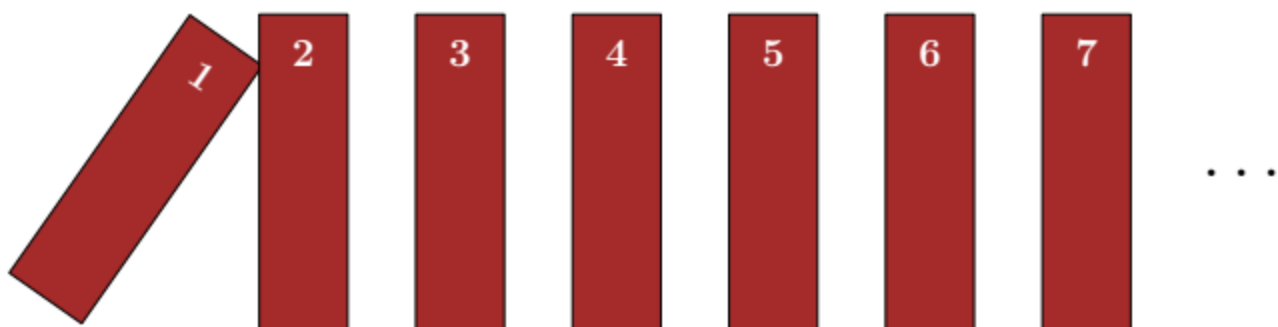
- **Fact 2:** If a domino falls, then the next domino also falls



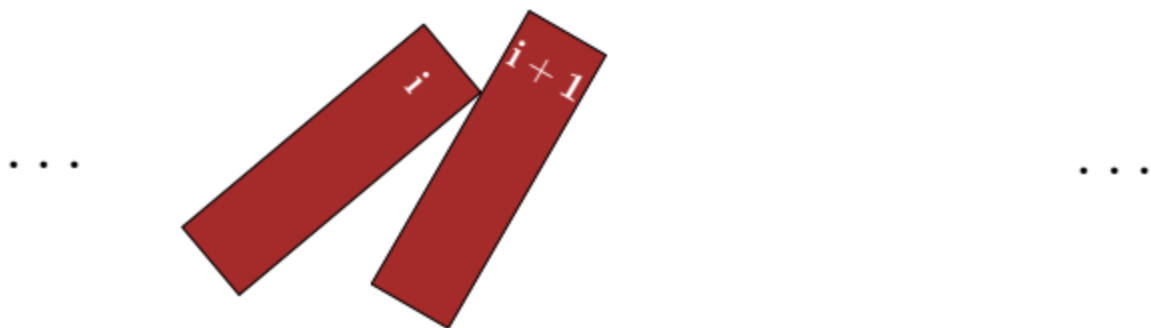
Principle of Induction

Suppose two rules/facts are given

1 Fact 1: First domino falls

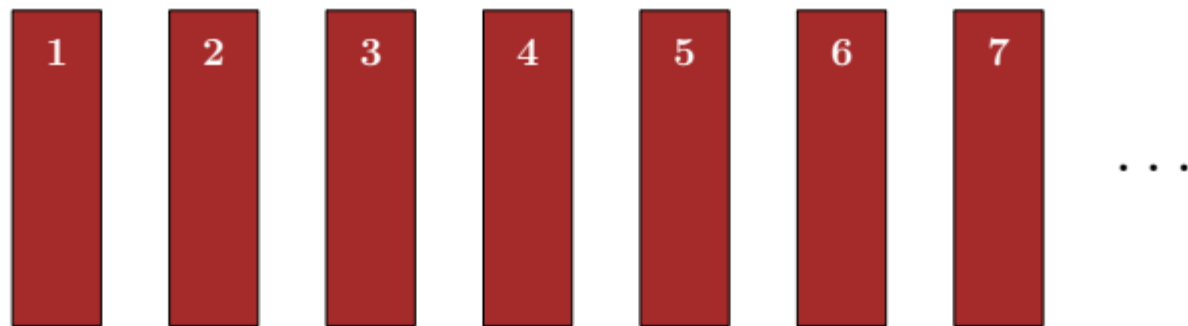


2 Fact 2: If a domino falls, then the next domino also falls



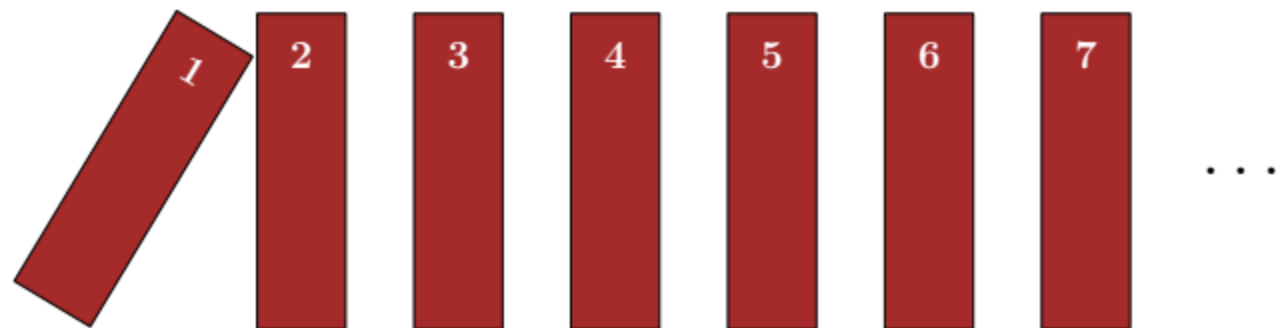
Principle of Induction

Applying both **Fact 1** and **Fact 2**



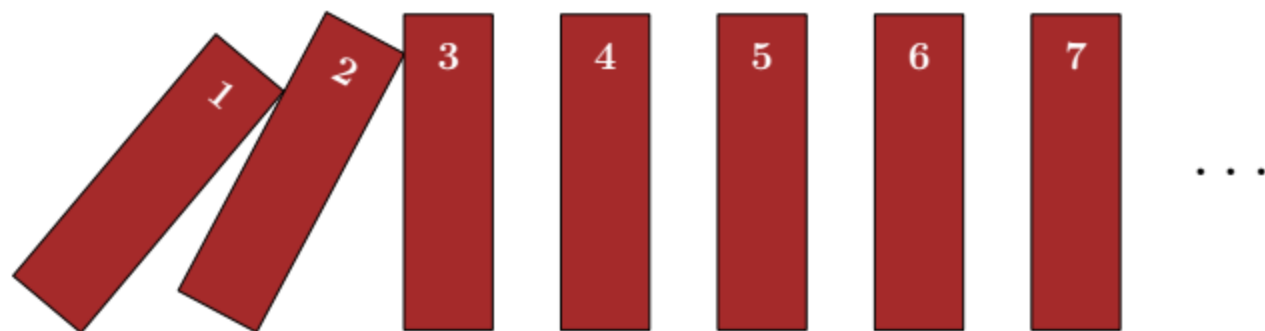
Principle of Induction

Applying both **Fact 1** and **Fact 2**



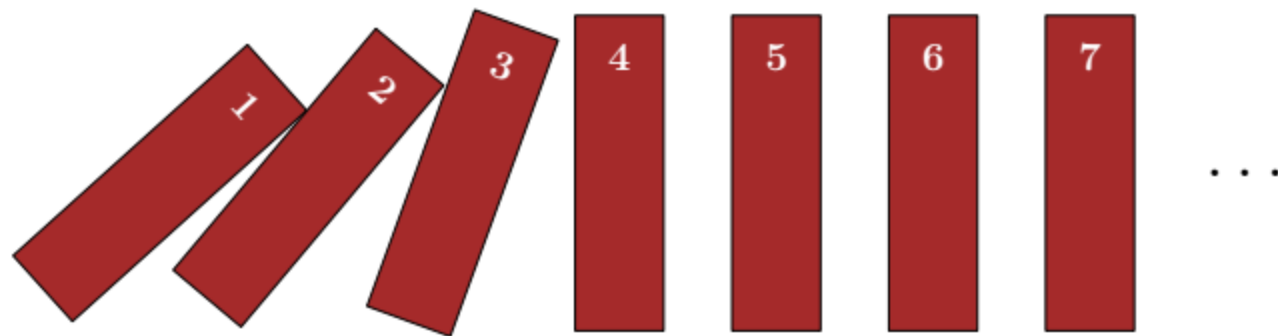
Principle of Induction

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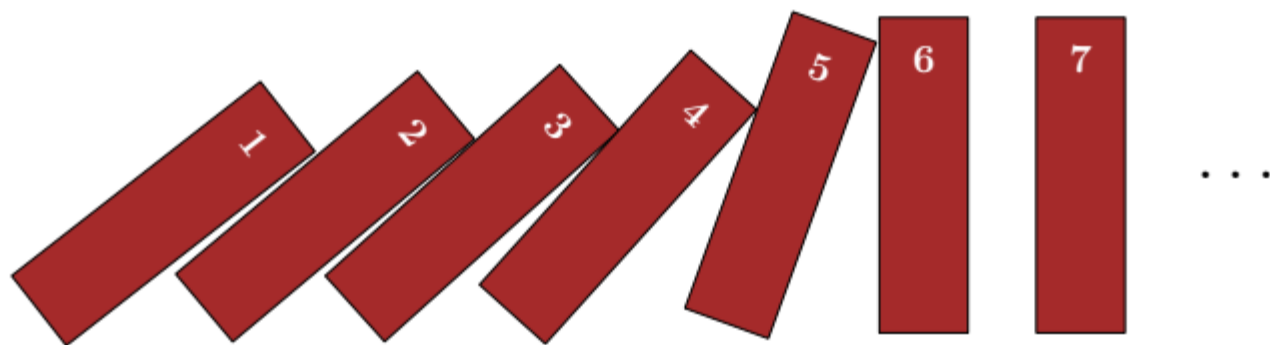
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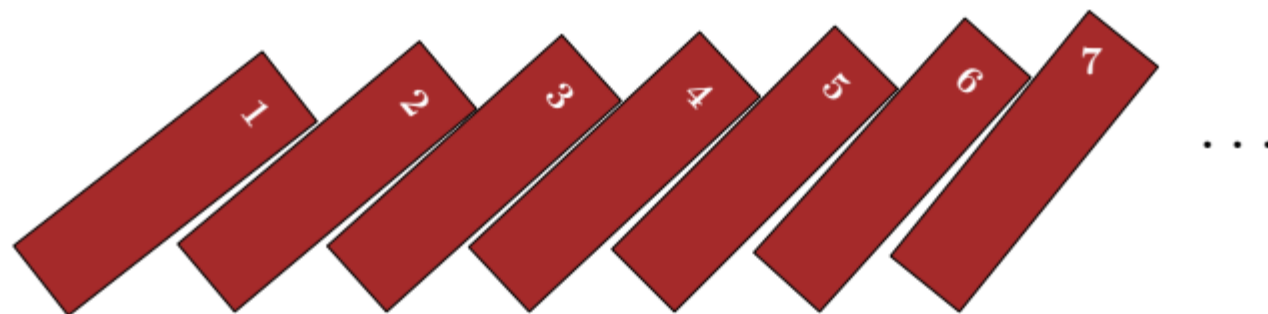
Principle of Induction

Applying both **Fact 1** and **Fact 2**



Principle of Induction

Applying both **Fact 1** and **Fact 2**



Can we conclude that all dominoes fall?

- Yes
- No

Proof by Induction

A proposition about non-negative integers, $\forall n P(n)$ is a sequence of propositions (dominoes)

$$P(0), P(1), P(2), \dots, P(n), P(n+1), \dots$$

Establish two facts

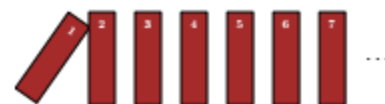
■ Prove $P(0)$

the first domino falls

■ Prove $\forall k \geq 0, P(k) \rightarrow P(k+1)$

if a domino falls, then the next domino also falls

Conclude that $P(n)$ is true for all n



Principle of Mathematical Induction

$$\left[P(0) \wedge \forall k \geq 0 [P(k) \rightarrow P(k+1)] \right] \longrightarrow \forall n \geq 0 P(n)$$

Proof by Induction

Theorem

$$P(n) : \sum_{i=0}^n i = \frac{n(n+1)}{2}$$

ICP 10-1

Formally rewrite to make a (explicit) statement about all positive integers

Theorem

$$\forall n \in \mathbb{N} \sum_{i=0}^n i = \frac{n(n+1)}{2} \quad \text{or} \quad \forall n \in \mathbb{N} P(n)$$

Proof by Induction

Theorem

$$\forall n \in \mathbb{N} \quad \sum_{i=0}^n i = \frac{n(n+1)}{2}$$

Proof Outline:

- 1 The proof is by induction on k
- 2 **Basis Step:** Prove that $P(k)$ is true for $k = 0$
- 3 **Induction Step:** Prove that for all k , $P(k) \rightarrow P(k+1)$
- 4 Conclude that $\forall n \in \mathbb{N} P(n)$ is true

Proof by Induction

Theorem

$$\forall n \in \mathbb{N} \quad \sum_{i=0}^n i = \frac{n(n+1)}{2}$$

Proof:

Basis Step: need to prove that $P(0)$ is true

$$P(0) : \sum_{i=0}^0 i = \frac{0(0+1)}{2}$$

This is clearly true, because both sides are equal to 0

Proof by Induction

Theorem

$$\forall n \in \mathbb{N} \quad \sum_{i=0}^n i = \frac{n(n+1)}{2}$$

Proof:

Inductive Step: need to prove that $\forall k \quad P(k) \rightarrow P(k+1)$ is true

Inductive Hypothesis (IH): Assume that $P(k)$ is true i.e. $\sum_{i=0}^k i = \frac{k(k+1)}{2}$

Given IH, show that $P(k+1)$ is true, i.e. $\sum_{i=0}^{k+1} i = \frac{(k+1)(k+2)}{2}$

Proof by Induction

Inductive Step: need to prove that $\forall k \ P(k) \rightarrow P(k+1)$ is true

Inductive Hypothesis (IH): Assume that $P(k)$ is true i.e. $\sum_{i=0}^k i = \frac{k(k+1)}{2}$

Given IH, show that $P(k+1)$ is true, i.e., $\sum_{i=0}^{k+1} i = \frac{(k+1)(k+2)}{2}$

$$\sum_{i=0}^{k+1} i = \sum_{i=0}^k i + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1) \quad \triangleright \text{by the Inductive Hypothesis}$$

$$= (k+1)\left(\frac{k}{2} + 1\right) = \frac{(k+1)(k+2)}{2} \implies P(k+1) \text{ is true} \quad \square$$

Proof by Induction

Theorem

The sum of first n positive odd integers is n^2

ICP 10-2

Restate it formally to make it a statement about all positive integers

Theorem

$$\forall n \in \mathbb{Z}^+ \quad \sum_{i=1}^n (2i - 1) = n^2$$

Proof by Induction

Theorem

$$\forall n \in \mathbb{Z}^+ \quad \sum_{i=1}^n (2i - 1) = n^2$$

ICP 10-3 What will be the basis step?

Proof: The proof is by induction on n .

Basis Step: $n = 1$

$$P(1) : \sum_{i=1}^1 (2i - 1) = 1 = 1^2$$

Clearly true!

Proof by Induction

Theorem

$$\forall n \in \mathbb{Z}^+ \quad \sum_{i=1}^n (2i - 1) = n^2$$

Inductive Hypothesis (IH): Assume $P(n)$ is true, i.e.

$$\sum_{i=1}^n (2i - 1) = n^2$$

Inductive Step: Given IH, show that $P(n + 1)$ is true,

$$\sum_{i=1}^{n+1} (2i - 1) = (n + 1)^2$$

Proof by Induction

Inductive Step: Given IH, show that $P(n+1)$ is true,

$$\sum_{i=1}^{n+1} (2i-1) = (n+1)^2$$

$$\sum_{i=1}^{n+1} (2i-1) = \sum_{i=1}^n (2i-1) + 2(n+1) - 1$$

$$= n^2 + 2(n+1) - 1 \quad \triangleright \text{by the inductive hypothesis}$$

$$= n^2 + (2n+1)$$

$$= (n+1)^2$$



Proof by Induction

A proposition about non-negative integers, $\forall n P(n)$ is a sequence of propositions (dominoes)

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Establish two facts

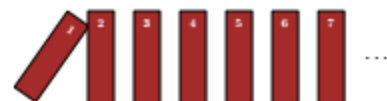
- Prove $P(0)$

the first domino falls

- Prove $\forall k \geq 0, P(k) \rightarrow P(k+1)$

if a domino falls, then the next domino also falls

Conclude that $P(n)$ is true for all n



Principle of Mathematical Induction

$$\left[P(0) \wedge \forall k \geq 0 \left[P(k) \rightarrow P(k+1) \right] \right] \longrightarrow \forall n \geq 0 P(n)$$

Proof by Induction

Theorem

Sum of first n powers of 2 is $2^n - 1$

Start with $2^0 = 1$

ICP 10-4

Formally, restate as a statement about all positive integers?

Theorem

$$\forall n \in \mathbb{N} \quad \sum_{i=0}^{n-1} 2^i = 2^n - 1$$

Proof by Induction

Theorem

$$\forall n \in \mathbb{N} \quad \sum_{i=0}^{n-1} 2^i = 2^n - 1$$

Proof: Basis Step: Rule: Sum of nothing is 0.

$$\sum_{i=0}^{-1} 2^i = 0 \quad \text{and} \quad 2^0 - 1 = 0$$

Inductive Step:

$$\sum_{i=0}^n 2^i = \sum_{i=0}^{n-1} 2^i + 2^n = 2^n - 1 + 2^n = 2^{n+1} - 1$$

Proof by Induction

ICP 10-5 Prove that

Theorem

The sum of squares of first n positive integers is $\frac{n(n+1)(2n+1)}{6}$

Proof by Induction

ICP 10-5 Prove that

Theorem

The sum of squares of first n positive integers is $\frac{n(n+1)(2n+1)}{6}$

Proof: Proof by induction on n

Basis Step: $n = 0$ $n = 1$ $1^2 = \frac{1(1+1)(2(1)+1)}{6}$

Inductive Step: Prove

$$\underbrace{\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}}_{\text{inductive hypothesis}} \implies \sum_{i=1}^{n+1} i^2 = \frac{(n+1)(n+2)(2n+3)}{6}$$

Proof by Induction

ICP 10-6

Prove that

Theorem

The sum of cubes of first n positive integers is $\left(\frac{n(n+1)}{2}\right)^2$

Proof by Induction

ICP 10-6

Prove that

Theorem

The sum of cubes of first n positive integers is $\left(\frac{n(n+1)}{2}\right)^2$

Proof: Proof by induction on n

Basis Step: $n = 1$ $1^3 = \left(\frac{1(1+1)}{2}\right)^2$

Inductive Step: Prove

$$\underbrace{\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2}_{\text{inductive hypothesis}} \Rightarrow \sum_{i=1}^{n+1} i^3 = \left(\frac{(n+1)(n+2)}{2}\right)^2$$

Proof by Induction

Theorem

An ATM with only \$3 and \$5 bills can generate any amount $n \geq 8$

Basis Step: $n = 8$ dollar. A \$3 bill and a \$5 bill

Induction Step: Need to prove that given that the ATM can generate n dollar it can generate $n + 1$ dollar amount

Case 1: Output of $\$n$ contains a \$5 bill

Remove a \$5 bill and add two \$3 bills

Case 2: Output of $\$n$ contains no \$5 bill, it must have $\geq$ three \$3 bills

Remove three of \$3 and add two \$5 bills

Proof by Induction

ICP 10-7 Prove that

Theorem

An ATM with only \$3 and \$7 bills can generate any amount $n \geq 12$

Basis Step: $n = 12$ dollar.

▷ $4 \times \$3$

Induction Step: Prove if ATM can generate $\$n$, then it can generate $\$n + 1$

Case 1: Output of $\$n$ contains $\geq$ two \$3 bills

Case 2: Output of $\$n$ contains $\leq$ one \$3 bill, it must have $\geq$ two \$7 bills

Well Ordering Principle

Principle of Well Ordering

Any nonempty set of nonnegative integers has a smallest element

What is the smallest element of

ICP 10-11 $\{x \in \mathbb{Z} : x < 7\}$

ICP 10-12 $\{x \in \mathbb{R} : 0 < x < 1\}$

ICP 10-13 $\{3.9 + \frac{1}{n} : n \in \mathbb{Z}^+\}$

ICP 10-14 $\{\lfloor 3.9 + \frac{1}{n} \rfloor : n \in \mathbb{Z}^+\}$

Why induction work

Principle of Well Ordering

Any nonempty set of nonnegative integers has a smallest element

Principle of Mathematical Induction

$$\left[P(0) \wedge \forall k \geq 0 [P(k) \rightarrow P(k+1)] \right] \longrightarrow \forall n \geq 0 P(n)$$

Proof using Mathematical Induction

- 1 *Basis Step*: Prove $P(0)$ is true
- 2 *Inductive Step*: $\underbrace{\text{Assume } P(n)}_{\text{IH:}}$ Using $P(n)$, prove $P(n+1)$

We argue why induction make sense using the [principle of well ordering](#) and how they are essentially equivalent

Why induction work

$$\left[\underbrace{P(0)}_{\text{basis step}} \wedge \underbrace{\forall k (P(k) \rightarrow P(k+1))}_{\text{induction step}} \right] \longrightarrow \forall n \geq 0 P(n)$$

Suppose we proved 1) $P(0)$ and 2) $\forall k P(k) \rightarrow P(k+1)$

and assume $\neg \forall n P(n) \equiv \exists n \neg P(n)$ $\triangleright$ i.e. induction does not work

Let $S = \{ k : \neg P(k) \}$ $\triangleright S \neq \emptyset$, why?

Let m be its least element $\triangleright$ one exists by well-ordering

■ $P(m)$ is false $\because m \in S$

■ $m \neq 0$ $\because P(0)$ is true by 1)

■ $P(m-1)$ is true $\because m$ is the least element of S

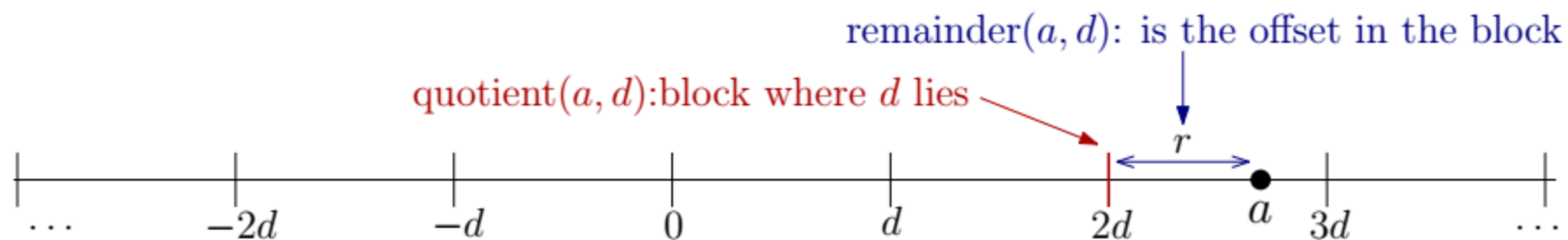
■ But $\underbrace{P(m-1)}_{\text{true}} \rightarrow \underbrace{P(m)}_{\text{false}}$ is true by 2) $\triangleright$ Contradiction

Proof using Principle of Well Ordering

Theorem (The Division Algorithm)

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$ such that $a = dq + r$

- q is called the **quotient**
- r is called the **remainder**
- d is called the **divisor**
- a is called the **dividend**



Proof using Principle of Well Ordering

Theorem (The Division Algorithm)

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$ such that $a = dq + r$

quotient and remainder when 11 is divided by 5

$$11 = 5 \times 2 + 1$$

Notation:

$$q = 2 = 11 \text{ div } 5$$

$$r = 1 = 11 \text{ mod } 5$$

Proof using Principle of Well Ordering

Theorem (The Division Algorithm)

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$ such that $a = dq + r$

quotient and remainder when -11 is divided by 5

$$-11 = 5 \times -3 + 4$$

Notation:

$$q = -3 = -11 \text{ div } 5$$

$$r = 4 = -11 \text{ mod } 5$$

Proof using Principle of Well Ordering

Theorem (The Division Algorithm)

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$ such that $a = dq + r$

$$11 = 5 \times 2 + 1$$

Given positive a , repeatedly subtract d until what remains r is $0 \leq r < d$

$$-11 = 5 \times -3 + 4$$

Given negative a , repeatedly add d until what remains r is $0 \leq r < d$